Nguyen Ba Thinh

AIOT Engineer | Edge AI | Computer Vision | Embedded Systems

Internet of Things and Artificial Intelligent Engineer

Hi, I'm Thinh (born August 4, 2003). AI Engineer with hands-on experience in Computer Vision and Edge AI systems. Experienced in deploying real-time AI models on embedded devices such as Jetson Nano and Hailo accelerators. Strong interest in building scalable and low-latency intelligent systems.



EXPERIENCE

Company: BEE HOOME TECHNOLOGY JSC (May 2025 - Present)

Role: Fullstack

Project build Chatbot for Page Facebook, Instagram

Result:

- Built an end-to-end automated chatbot on n8n that auto-replies to both Facebook Messenger and Page comments 24/7 for a premium kitchen-cabinet brand
- Integrated a RAG pipeline using PostgreSQL + PGVector and Google Gemini embeddings, so the bot answers only from the company knowledge base instead of hallucinating
- Designed an AI sales-consultant agent (LangChain + Google Gemini) with a controlled Canada persona and strict answer rules to consult products and qualify leads
- Automated knowledge-base sync: a Google Drive trigger detects new files, splits them into ~800-token chunks, embeds and upserts them into the vector store with no manual work
- Implemented per-user bot ON/OFF control for smooth human handover, plus echo/duplicate filtering
- Used PostgreSQL chat memory for multi-turn context and n8n Data Tables to track message status per customer
- Self-hosted the entire stack with Docker — deployed n8n and a PostgreSQL + PGVector database in containers, owning data, cost and scaling instead of relying on cloud SaaS

Technical Stack: n8n (self-hosted), Webhooks · Google Gemini, LangChain AI Agent, RAG · PostgreSQL + PGVector, Gemini Embeddings · Docker, Docker Compose, Linux (VPS) · Facebook Graph API v19.0, Google Drive API · JavaScript, JSON

Company: HBQ Technology (April 2025 - April 2026)

Role: R&D

Project 14G/5G Drone Streaming & Control System(04/2025-07/2025) -(02/2026-04/2026)

R*esult:*

- Developed a 4G/5G-enabled drone system on a custom embedded platform
- Implemented real-time video & telemetry streaming via RTSP, achieving <300ms latency
- Customized and integrated QGroundControl for monitoring and control, improving operational efficiency by ~40%
- Designed TCP/UDP-based communication and authentication, reducing packet loss by ~5% in unstable networks
- Deployed computer vision for landing zone detection, achieving ~80% accuracy in real-world conditions

Technical Stack

- Language: Go, Python,C++
- Computer Vision & Streaming: OpenCV,GStreamer,RTSP / MediaMTX, IMX219 cameras
- Backend & API: Go REST framework, websocket, gomavlib
- Frontend:React, TypeScript
- Networking & DevOps:WireGuard VPN, Git, Makefile, git, Linux
- Tool :PX4 Autopilot, MAVLink protocol, Qt 6 Framework.
- Hardware: Compute module 5 (customer board), SIM7600G-H-M2.
- AI tool: Copilot Github.

AREAS OF EXPERTISE

I focus on building AI systems, especially Computer Vision models on edge devices.
I also have experience in device communication and system integration, where edge devices communicate with servers to process data and execute real-time tasks

EDUCATION

Bachelor of Engineer
Industrial University of Ho Chi Minh
2020-2026
Completed degree requirements, awaiting graduation

CERTIFICATE

Engineering Project (eProjects)
ASU and Dow Vietnam STEM Program

RESEARCH

Efficient Face Recognition on Embedded Devices via Structured Pruning and Stereo-Camera Feature Fusion

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Project: Edge AI Face Recognition System(05/2025-01/2026)

Result:

- Built an edge-based face recognition system on Hailo-8L AI accelerator and Raspberry Pi 5
- Achieved ~25 FPS real-time inference under edge constraints
- Applied quantization and pruning, reducing model size by ~30% and speed detect about 3s for video
- Developed custom inference modules, improving recognition accuracy by ~6%
- Enhanced robustness using advanced pre-processing and post-processing techniques

Technical Stack:

- Language: Python, QML, Shell Script, Jupyter Notebook
- AI / ML Framework: PyTorch, SCRFD, ONNX
- Edge AI & Hardware Acceleration:Hailo Runtime, hailo_platform SDK
- Computer Vision & 3D Processing: OpenCV, Picamera2, V4L2, NumPy,Matplotlib
- Database & Storage: CSV, JSON, Remote HTTP API
- Hardware Integration: Hailo M.2 AI Accelerator, Dual OV5647 Camera, Raspberry Pi 5
- AI tool: Copilot Github

Personal project: Detect muscle group from video

Result:

- Converted and optimized models across PyTorch → ONNX → TensorRT pipeline
- Reduced inference latency by ~30% through TensorRT optimization
- Built deployment pipelines for scalable edge/embedded AI systems

Technical Stack:

- Language: Python
- AI / ML Framework: PyTorch, TensorRT, ONNX
- Edge AI & Hardware Acceleration:Hailo Runtime, hailo_platform SDK
- Computer Vision & 3D Processing: OpenCV,YOLO
- Hardware Integration: JetsonNano B01, OV5647 Camera